

Sentencing Disparities between Sex and Race

Kristian A. Nunez

University of Central Florida

15 July 2025

Data Preparation: Filtering

- Imported active and released inmate datasets
- Filtered for homicide-related charges using pattern matching.
- Dropped rows with missing race or sex information.
- Merged both datasets into a single unified table.

Data Cleaning: Standardization

- Mapped race and sex codes to readable labels.
- Categorized 35+ charge descriptions into 10 homicide types.
- Created sentence type: life, death, or fixed-term.
- Converted sentence terms into total days.

Categorizing the Charges

```
charge_categories = {
    '1ST DG MUR/PREMED. OR ATT.': 'FIRST_DEGREE_MURDER',
    '2ND DEG.MURD,DANGEROUS ACT': 'SECOND_DEGREE_MURDER',
    'HOMICIDE,MANSL.CUL.NEGLI': 'NEGLIGENT_MANSLAUGHTER',
    'a-s, murder o/t 782.04(4) a-s': 'THIRD_DEGREE_MURDER',
    '1ST DEG MUR,COM.OF FELONY': 'FIRST_DEGREE_MURDER',
    'AT.FLNY.MURD/782.04(3) OFF.': 'ATTEMPTED_MURDER',
    'AT.FLNY.MURD/NOT 782.04(3)': 'ATTEMPTED_MURDER',
    'SECOND DEG.MURDER,COMM.OF FELO': 'SECOND_DEGREE_MURDER',
    'DUI MANSLAUGHTER': 'DUI_BUI_MANSLAUGHTER',
    'HOMICIDE-NEGLIG MANSL-VEH': 'VEHICULAR_MANSLAUGHTER',
    'DUI,MANSLAUGHTER': 'DUI_BUI_MANSLAUGHTER',
    'HOMICIDE NEGLIG MANSL VEH': 'VEHICULAR_MANSLAUGHTER',
    '1ST DEG MUR,DEATH FRM DRUGS': 'GENERAL_HOMICIDE',
    'ATTEMPT MURDER LAW ENFORCE OFF': 'ATTEMPTED_MURDER',
    'HOMICIDE,MANSL.UNNC.KILL 'G': 'GENERAL_MANSLAUGHTER',
    'HOMICIDE-OTHER/OTHER STATE': 'GENERAL_HOMICIDE',
    'HOMICIDE-WLFL KILL-NONFMLY-GUN': 'GUN_RELATED_HOMICIDE',
    'HOMICIDE,KILL UNBORN CHLD': 'FETAL_HOMICIDE',
    'MANSLAUGHTER/UNBORN CHILD': 'FETAL_HOMICIDE',
    'BUI MANSLAUGHTER': 'DUI_BUI_MANSLAUGHTER',
    'FELONY MURDER-NONSEX': 'GENERAL_HOMICIDE',
    'HOMICIDE,MANSL.ASST.SELF-MUR': 'GENERAL_HOMICIDE',
    'AT.FLNY.MURD LEO/NOT 782.04(3)': 'LAW_ENFORCEMENT_KILL',
    'HOMICIDE-WILFUL W/VESSEL': 'VEHICULAR_MANSLAUGHTER',
    '2ND DG.MURDER/UNBORN CHILD': 'FETAL_HOMICIDE',
    '1ST DG.MURDER/UNBORN CHILD': 'FETAL_HOMICIDE',
    '2ND DEG ATT MUR/V - LEO': 'ATTEMPTED_MURDER',
    '1ST DEG ATT MUR/V - LEO': 'ATTEMPTED_MURDER',
    '2nd DEG, DA MUR/V - LEO': 'SECOND_DEGREE_MURDER',
    '1st DEG MUR/V - LEO': 'FIRST_DEGREE_MURDER',
    'AT.FLNY.MURD LEO/782.04(3) OFF': 'ATTEMPTED_MURDER',
    'HOMICIDE': 'GENERAL_HOMICIDE'
```

Feature Engineering & Export

- Calculated age at offense and flagged minors.
- Classified sentencing system by law in effect at time of offense.
- Removed unnecessary columns and saved cleaned data:
 - `fixed_cases.csv`, `life_cases.csv`, `death_cases.csv`

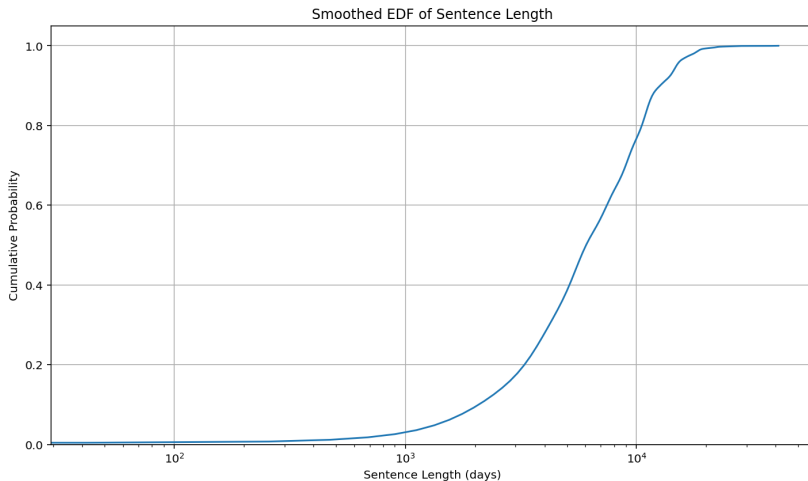
Dataset Focus for Analysis

- This analysis uses the `fixed_cases.csv` dataset Post 1998, so with CPC Guidelines.
- Although it contains many variables, the key columns used are:
 - **Race**
 - **Sex**
 - **Charge**
 - **Sentence Days**
- Other columns were dropped or are not part of this specific analysis.

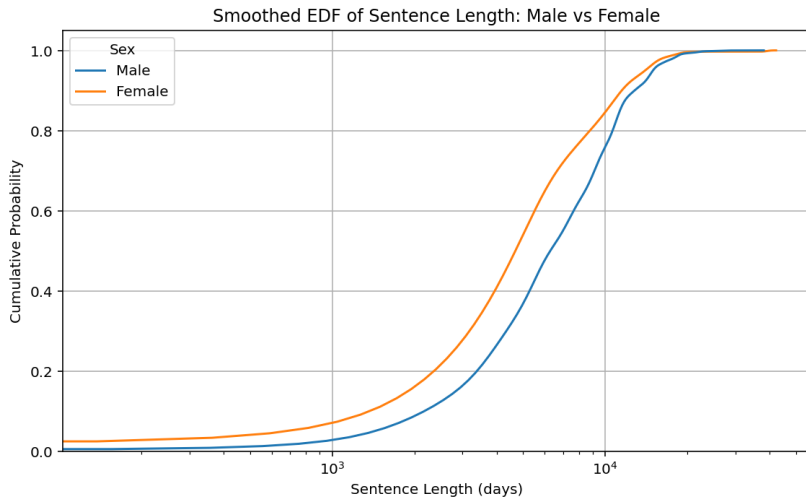
Fixed Cases Sample Cleaned Data Rows

Charge	Sex	Race	Sentence Days
FIRST_DEGREE_MURDER	Male	White	10795
ATTEMPTED_MURDER	Female	Black	6935
DUI_BUI_MANSLAUGHTER	Male	Hispanic	3995
GUN_RELATED_HOMICIDE	Male	Asian	7990

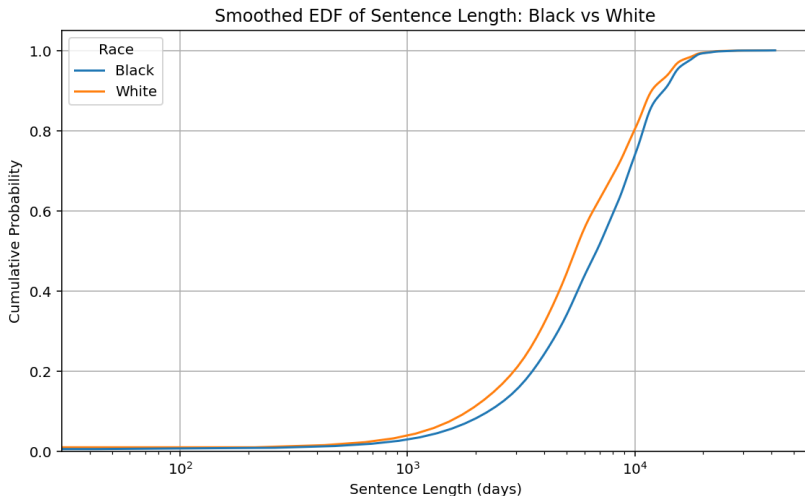
Smooth Empirical Distribution Function for All



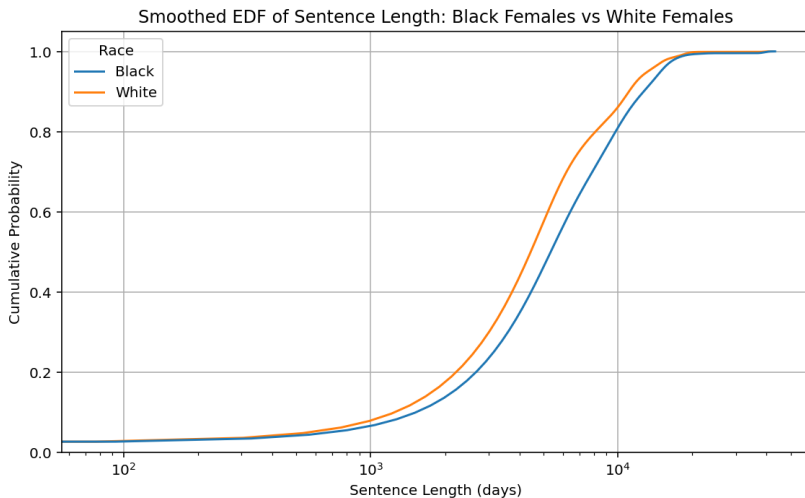
Smooth Empirical Distribution Function: Male vs Female



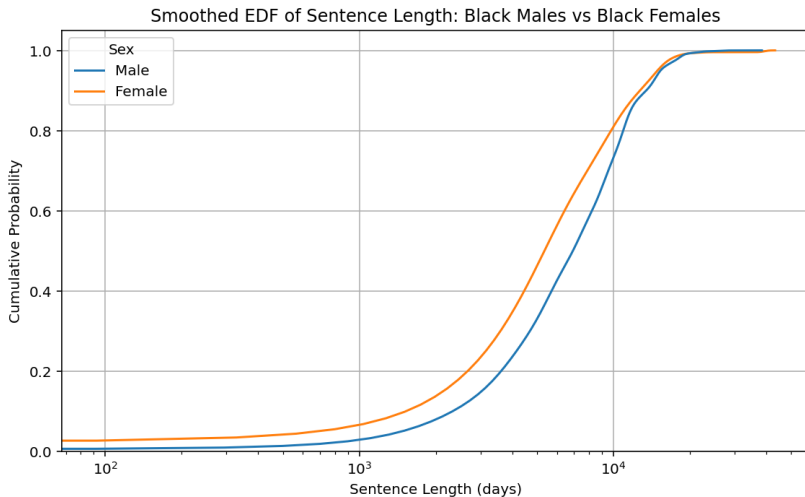
Smooth Empirical Distribution Function: Black vs White



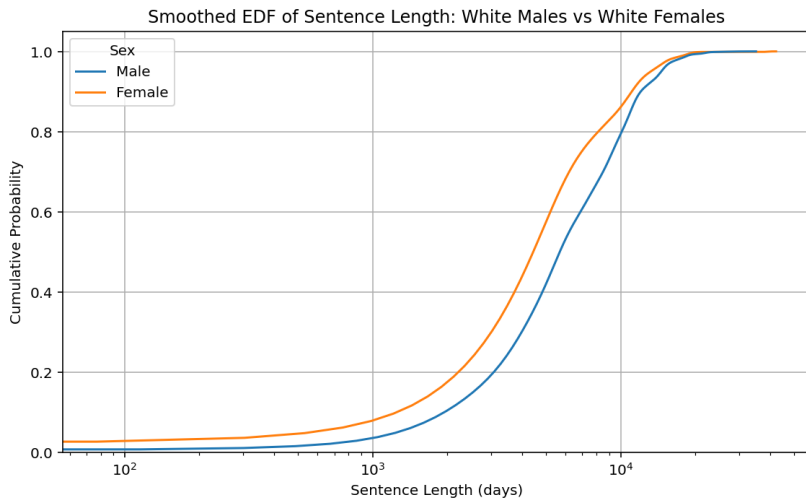
Smooth Empirical Distribution Function: Black Vs White Females



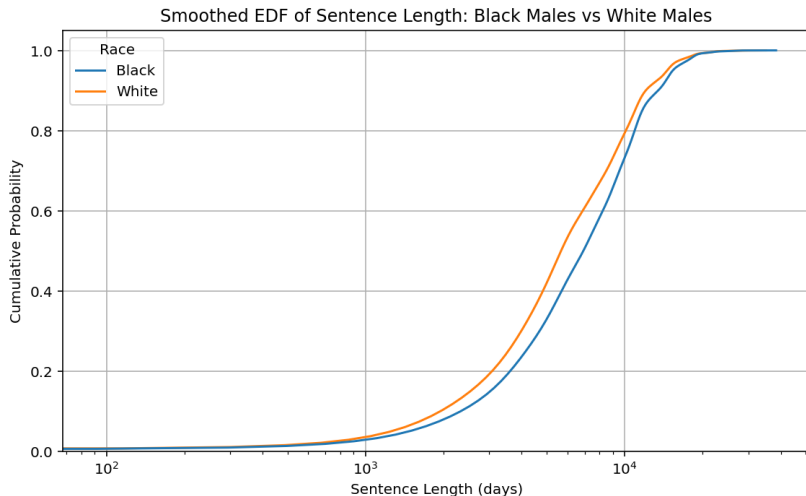
Empirical Distribution Function: Black Males vs Black Females



Empirical Distribution Function: White Males vs White Females



Empirical Distribution Function: Black Vs White Males

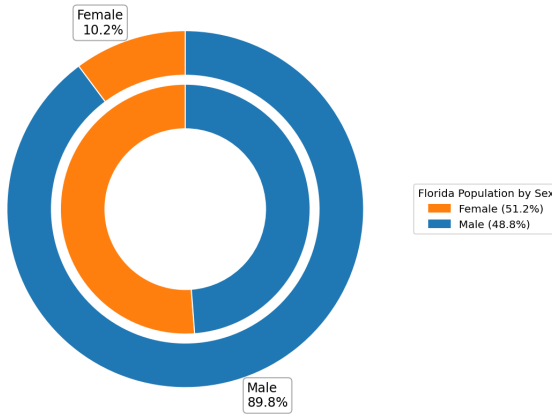


EDF Summary: Sentencing by Race and Sex

- **Overall Pattern:** Black individuals and men tend to receive longer sentences on average.
- **Distribution Differences:**
 - Most disparities appear around the **50th (median)** and **75th percentiles**.
 - Indicates systemic differences in typical (not extreme) sentencing outcomes.
- **Top-End Similarity:**
 - At the highest quantiles, EDF curves tends to converge.
 - Suggests disparities are smaller for the most severe crimes (e.g., First Degree or Second Degree Murder).

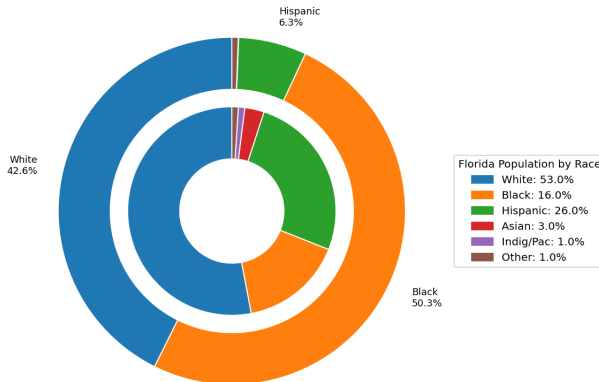
Sex Representation in Homicide Cases vs. Florida Population

Sex Donut Chart
Outer = Homicide Convictions | Inner = Florida Population



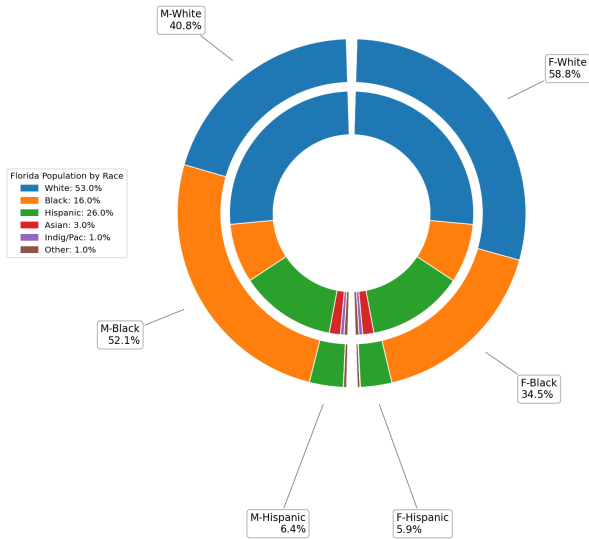
Race Representation in Homicide Cases vs. Florida Population

Race Donut Chart
Outer = Homicide Convictions | Inner = Florida Population



Intersectional Representation: Race and Sex Combined

Sex x Race Donut Chart
Outer = Homicide Convictions | Inner = Florida Population
Left = Male | Right = Female



Summary: Demographic Over/Underrepresentation

- **By Sex:**

- **Men** are severely overrepresented in homicide convictions.
- **Women** are significantly underrepresented.

- **By Race:**

- **Black individuals** are heavily overrepresented.
- **White individuals** are slightly underrepresented.
- **Hispanic, Asian, and Other groups** are extremely underrepresented.

- **By Race and Sex Combined:**

- **Black males and females** are strongly overrepresented.
- **White women** are slightly overrepresented.
- **White men** are slightly underrepresented.
- All other race-sex groups are **very underrepresented** regardless of sex.

Summary Statistics by Subgroup

- Prior to calculating ARO, computed descriptive statistics for each subgroup.
- Grouped data in three ways:
 - ① **By Sex**
 - ② **By Race and Sex**
 - ③ **By Charge, Race, and Sex**
- For each group, calculated:
 - **Count** of unique individuals
 - **Average sentence length (days)**
 - **Standard deviation** of sentence lengths
 - **Standard error of the mean**
- Used these values to understand the distribution of sentencing and to support the ARO calculations. I also plan on making a tree with this, to understand patterns better.

Summary Statistics By Sex

Sex	count	percentage	average_sentence	median_sentence	std_sentence	sem_sentence
Female	1215	10.198942331906322	5781.068312757202	4745.0	4320.643168735463	123.95391879568355
Male	10698	89.80105766809368	7250.2280800149565	6570.0	4198.766025379105	40.594797340587476

Summary Statistics By Sex and Race

Sex	Race	count	percentage	average_sentence	median_sentence	std_sentence	sem_sentence
Female	Asian	1	0.008394191219675984	2555.0	2555.0	nan	nan
Female	Black	419	3.5171661210442373	6473.510739856802	5475.0	4651.9141994862675	227.26088023126462
Female	Hispanic	72	0.6043817678166709	5261.097222222223	4380.0	5266.759046089883	620.6935060609585
Female	Indigenous / Pacific Islander	3	0.025182573659027953	3650.0	3285.0	965.6992285385755	557.5467095529606
Female	Other / Unknown	6	0.05036514731805591	6265.833333333333	4562.5	3725.26565048275	1520.8333333333335
Female	White	714	5.993452530848653	5436.551820728291	4380.0	3971.4960984853305	148.6295060088018
Male	Asian	2	0.01678838243935197	12775.0	12775.0	2580.9397513308986	1825.0
Male	Black	5576	46.80601024091329	7660.581958393113	7300.0	4232.750461924581	56.68410361598747
Male	Hispanic	683	5.733232603038697	7034.799414348463	5475.0	4147.369145166042	158.6946699845834
Male	Indigenous / Pacific Islander	9	0.07554772097708386	8491.444444444445	9125.0	4747.589127944601	1582.5297093148667
Male	Other / Unknown	65	0.545622429278939	8062.815384615385	7300.0	4135.445354237511	512.9388669210249
Male	White	4363	36.623856291446316	6742.312858125143	5475.0	4104.189424164573	62.13478364327486

Summary Statistics By Sex, Race and Charge: NOTE JUST SAMPLE MUCH BIGGER

Sex	Race	charge	count	percentage	average_sentence	median_sentence	std_sentence	sem_sentence
Female	Asian	NEGLIGENT_MANSLAUGHTER	1	0.000394191219675984	2555.0	2555.0	nan	nan
Female	Black	ATTEMPTED_MURDER	73	0.6127759590363469	6519.534246575343	4380.0	6784.931746733983	794.1161952851916
Female	Black	DUI_BUI_MANSLAUGHTER	46	0.38613279610509527	3821.1739130434785	3650.0	2199.5346609311887	324.3036930381903
Female	Black	FIRST_DEGREE_MURDER	2	0.0167883243935197	10037.5	10037.5	3871.4096269963475	2737.4999999999995
Female	Black	GENERAL_HOMICIDE	1	0.000394191219675984	6570.0	6570.0	nan	nan
Female	Black	NEGLIGENT_MANSLAUGHTER	83	0.6967178712331067	4268.361445783133	3799.0	2252.60735960447	247.255779874315
Female	Black	SECOND_DEGREE_MURDER	193	1.620078905397465	8261.487046632124	7778.0	4221.702581735648	303.8848013309746
Female	Black	THIRD_DEGREE_MURDER	7	0.05875933853773189	4927.142857142857	2920.0	4483.9573826261415	1694.7765891334554
Female	Black	VEHICULAR_MANSLAUGHTER	14	0.11751867707546378	3630.3571428571427	3512.5	1754.9429798820713	469.6282402887731
Female	Hispanic	ATTEMPTED_MURDER	15	0.12591286629513976	6372.866666666667	3650.0	9469.14016210151	2444.9214767148633
Female	Hispanic	DUI_BUI_MANSLAUGHTER	12	0.10073829463611181	3494.0833333333333	3102.5	1860.079688338566	536.9587543882132
Female	Hispanic	GENERAL_HOMICIDE	2	0.0167883243935197	1915.0	1915.0	1159.655121145938	820.0
Female	Hispanic	GENERAL_MANSLAUGHTER	1	0.000394191219675984	366.0	366.0	nan	nan
Female	Hispanic	NEGLIGENT_MANSLAUGHTER	9	0.07554772097708386	5312.7777777777777	5475.0	2065.64776630858736	688.5492554352912
Female	Hispanic	SECOND_DEGREE_MURDER	26	0.2182489717115756	6352.346153846154	5475.0	4289.0562499674215	841.1531351388841
Female	Hispanic	VEHICULAR_MANSLAUGHTER	7	0.05875933853773189	3443.5714285714284	3650.0	2009.7775877894926	759.6245268346114
Female	Indigenous / Pacific Islander	DUI_BUI_MANSLAUGHTER	3	0.025182573659027953	3650.0	3285.0	965.6992285385755	557.546709529606
Female	Other / Unknown	ATTEMPTED_MURDER	1	0.000394191219675984	10950.0	10950.0	nan	nan
Female	Other / Unknown	DUI_BUI_MANSLAUGHTER	2	0.0167883243935197	4197.5	4197.5	1806.6578259316288	1277.4999999999998
Female	Other / Unknown	NEGLIGENT_MANSLAUGHTER	1	0.000394191219675984	3650.0	3650.0	nan	nan
Female	Other / Unknown	SECOND_DEGREE_MURDER	2	0.0167883243935197	7300.0	7300.0	5161.879502661797	3650.0
Female	White	ATTEMPTED_MURDER	79	0.6631411063544027	5645.936708860676	5475.0	3839.26152351643	431.9506690663927
Female	White	DUI_BUI_MANSLAUGHTER	254	2.1321245697977	3757.236220472441	3799.5	1747.2923319882096	1099.4384717621947
Female	White	FIRST_DEGREE_MURDER	4	0.0335767648780394	3692.25	1726.5	4883.731658953701	2441.8658294768506
Female	White	GENERAL_HOMICIDE	1	0.000394191219675984	366.0	366.0	nan	nan
Female	White	GENERAL_MANSLAUGHTER	1	0.000394191219675984	5475.0	5475.0	nan	nan
Female	White	NEGLIGENT_MANSLAUGHTER	110	0.9233610341643582	4111.5545454545454	3650.0	2469.6982788876485	235.47649156436012
Female	White	SECOND_DEGREE_MURDER	206	1.7292033912532527	8855.432038834952	9125.0	4805.762261848214	334.8330857105234
Female	White	THIRD_DEGREE_MURDER	13	0.1091244858557878	3328.076923076923	3285.0	2128.809140459464	588.2066213485622
Female	White	VEHICULAR_MANSLAUGHTER	46	0.38613279610509527	3064.478260869565	3102.5	1657.840760871139	244.4352847761638
Male	Asian	SECOND_DEGREE_MURDER	2	0.0167883243935197	1275.0	1275.0	2580.939751308086	1825.0
Male	Black	ATTEMPTED_MURDER	1325	1.122303366070678	7532.027924528302	7300.0	4429.7724328255845	121.69520790745551
Male	Black	DUI_BUI_MANSLAUGHTER	148	1.242340005120457	4388.263513513513	4204.0	2046.9489801573357	168.25816970180009
Male	Black	FETAL_HOMICIDE	4	0.0335767648780394	12775.0	13687.5	6495.222603319048	3247.6113016595236
Male	Black	FIRST_DEGREE_MURDER	78	0.6547469151347267	9362.551282051281	9125.0	4692.708761825095	531.3446358943678
Male	Black	GENERAL_HOMICIDE	9	0.07554772097708386	9033.333333333334	7300.0	5742.735004434942	1914.2450281449808
Male	Black	GENERAL_MANSLAUGHTER	2	0.0167883243935197	5657.5	5657.5	2322.8457761978004	1642.4999999999998
Male	Black	NEGLIGENT_MANSLAUGHTER	918	7.705867539662553	5456.742919389978	5475.0	3049.839391306248	108.6332953913658
Male	Black	SECOND_DEGREE_MURDER	2851	23.931839167296232	8793.68092799018	9125.0	4145.1999142386995	77.63301495737413
Male	Black	THIRD_DEGREE_MURDER	112	0.9401494166637102	5248.5842151825715	5475.0	2479.5063015191911	234.2913214442092
Male	Black	VEHICULAR_MANSLAUGHTER	129	1.082850667338202	4217.472866217054	4815.0	2427.906415764668	213.76524042337996
Male	Hispanic	ATTEMPTED_MURDER	137	1.1500041970956099	7123.248175182482	7300.0	3796.884606801738	324.3897439287955

Charge-Level ARO by Demographic Sub-Groups

- For each **charge type**, calculated **ARO (Achievement Relative Opportunity)**:
 - Based on observed vs. expected subgroup counts
 - Expected counts derived from Florida's population proportions
- ARO computed in three ways:
 - 1 **By Sex** – compares Male vs. Female, regardless of race
 - 2 **By Race** – compares racial groups, regardless of sex
 - 3 **By Race within Sex** – calculates ARO separately for males and females across races
- Example: For **First Degree Murder**, calculated how over- or under-represented each group is compared to population expectations

What is ARO? (Achievement Relative Opportunity)

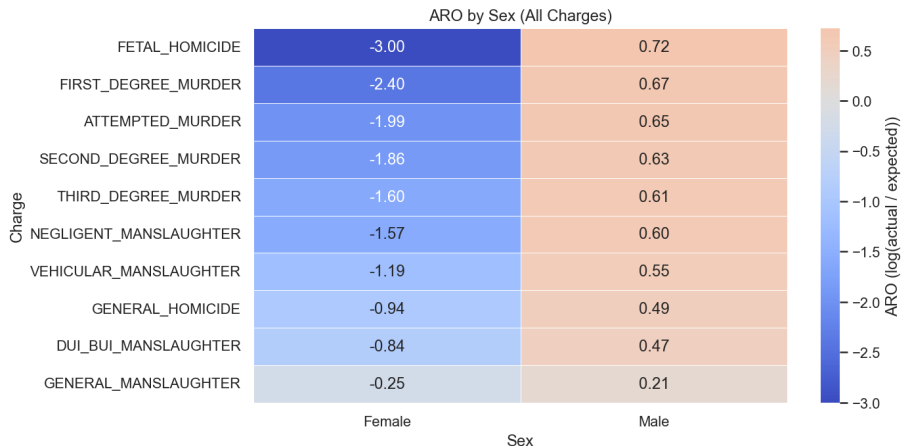
- ARO measures **over- or under-representation** of a group in a specific outcome — in this case, specific **homicide charges**.
- It compares the **observed number** of individuals from a group charged with a crime to the **expected number** based on that group's share of the Florida population.

ARO Formula

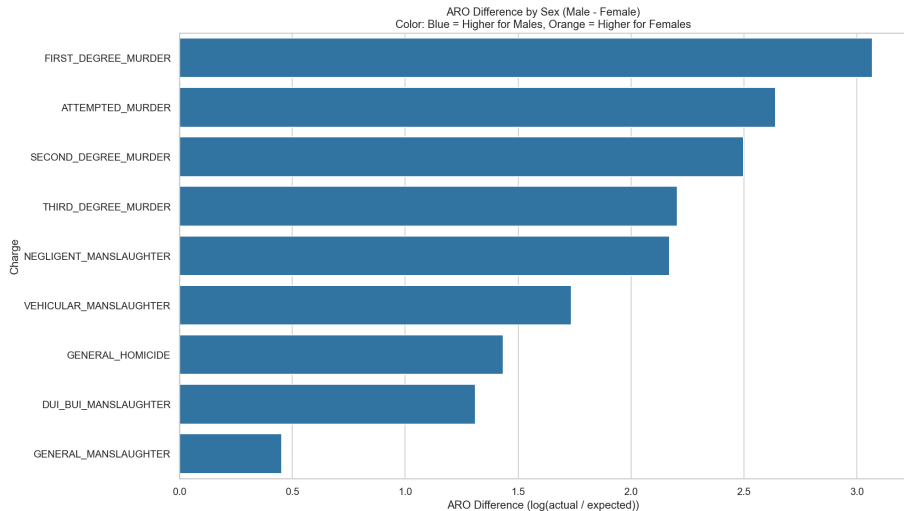
$$\text{ARO} = \log \left(\frac{\text{Observed Count}}{\text{Expected Count}} \right)$$

- Interpretation:
 - $\text{ARO} > 0$: Group is **over-represented**
 - $\text{ARO} < 0$: Group is **under-represented**
 - $\text{ARO} \approx 0$: Representation is roughly proportional
- ARO highlights disparities in how often different demographic groups are charged with specific homicide offenses.

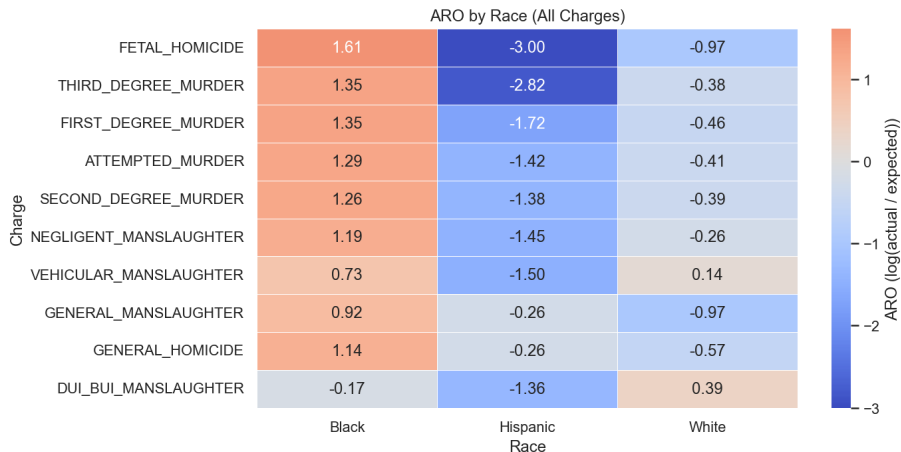
ARO Heatmap: Charge and Sex



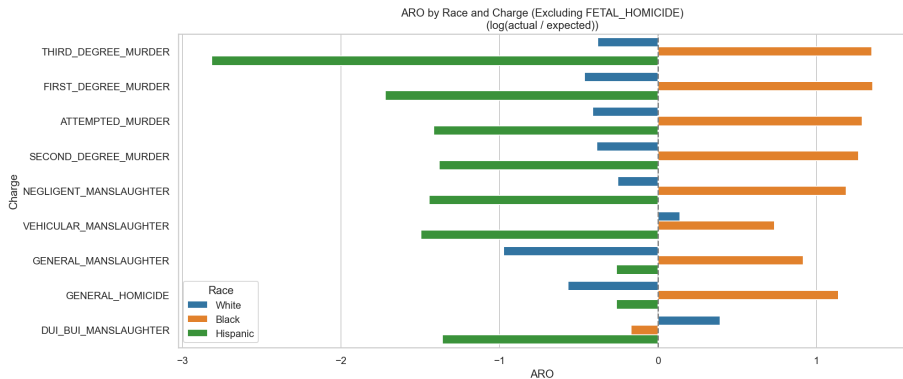
ARO Bar Plot: Charge and Sex



ARO Heatmap: Charge and Race



ARO Bar Plot: Charge and Race



ARO Summary by Sex and Race

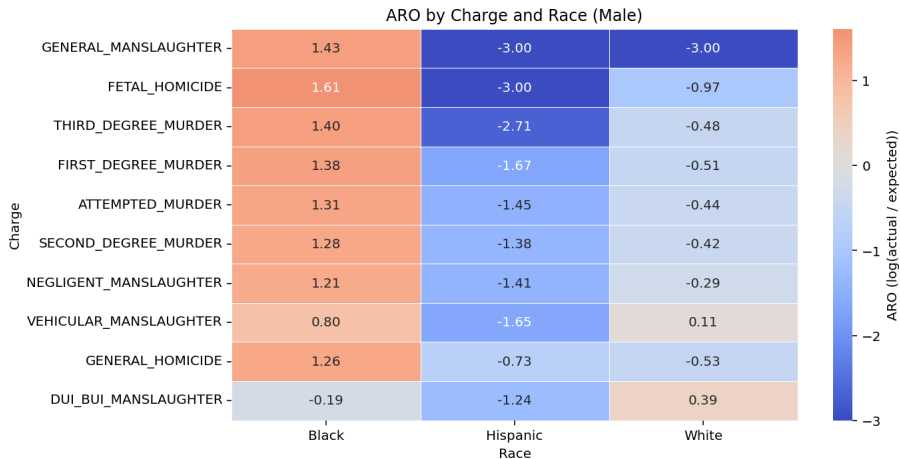
- **By Sex:**

- Men are consistently **overrepresented** across all homicide categories.
- Women are consistently **underrepresented** in every category.

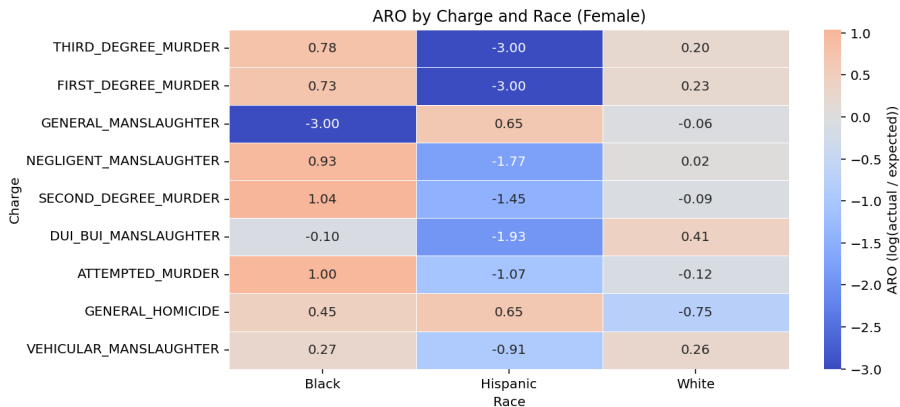
- **By Race:**

- **Black individuals** are overrepresented in all categories **except** DUI/BUI Manslaughter.
- **White individuals** are underrepresented in most categories, but slightly overrepresented in:
 - Vehicular Manslaughter
 - DUI/BUI Manslaughter
- **Hispanic individuals** are underrepresented across all homicide categories — often significantly.

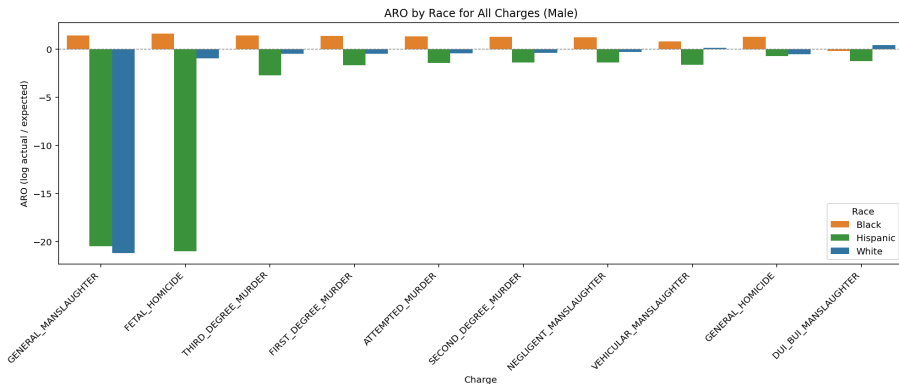
ARO Heatmap — Male (Race within Charge)



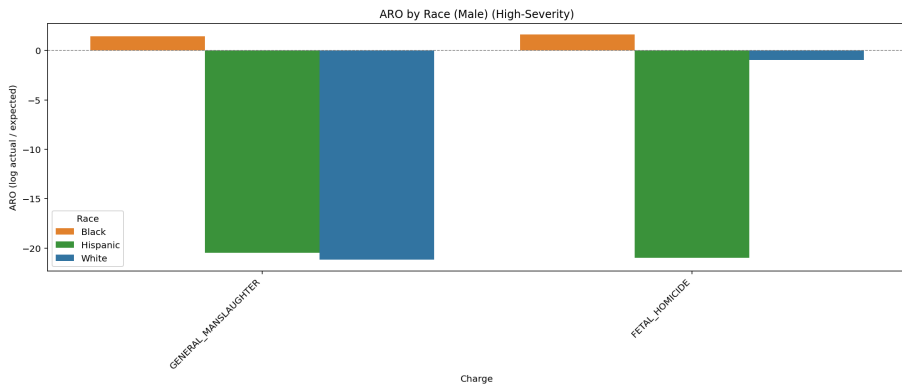
ARO Heatmap — Female (Race within Charge)



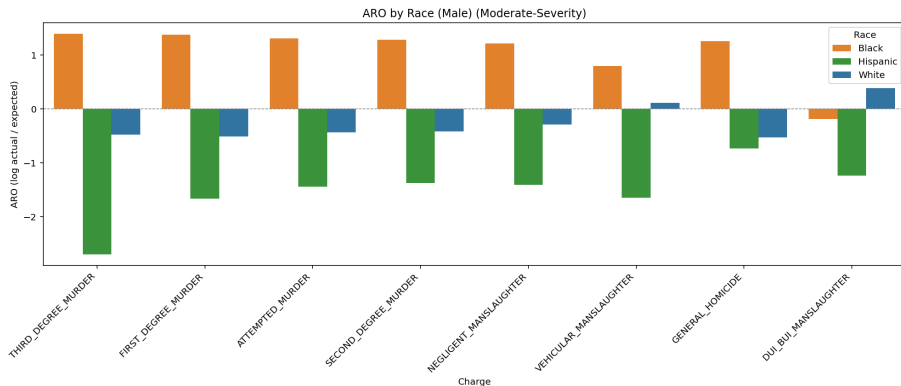
ARO Bar Plot — Male (All Charges)



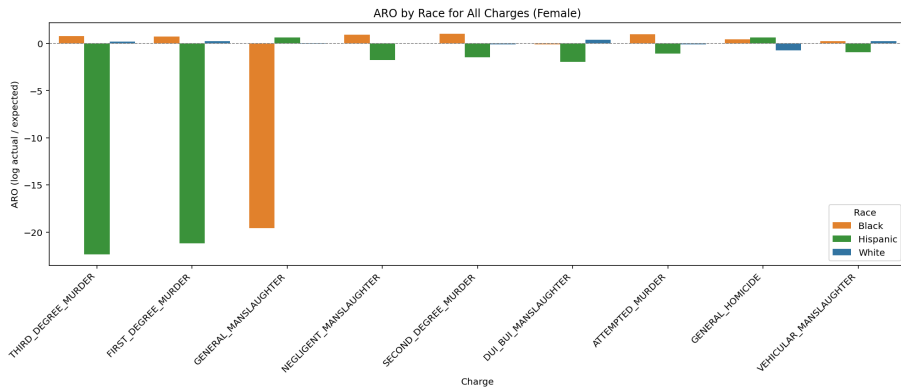
ARO Bar Plot — Male (Severe Disparities)



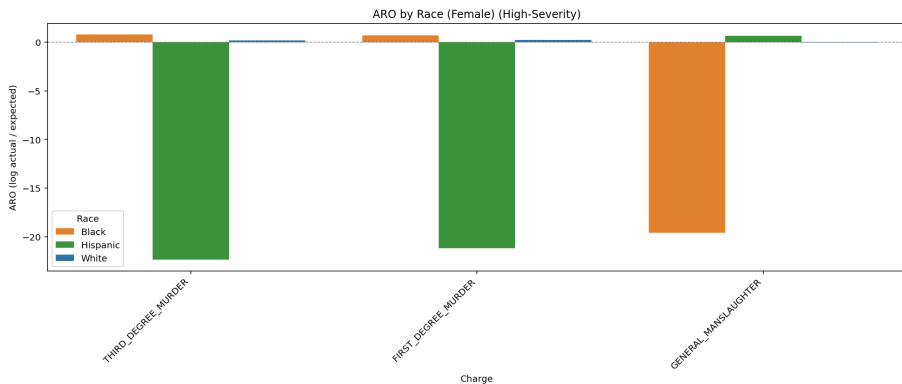
ARO Bar Plot — Male (Moderate Disparities)



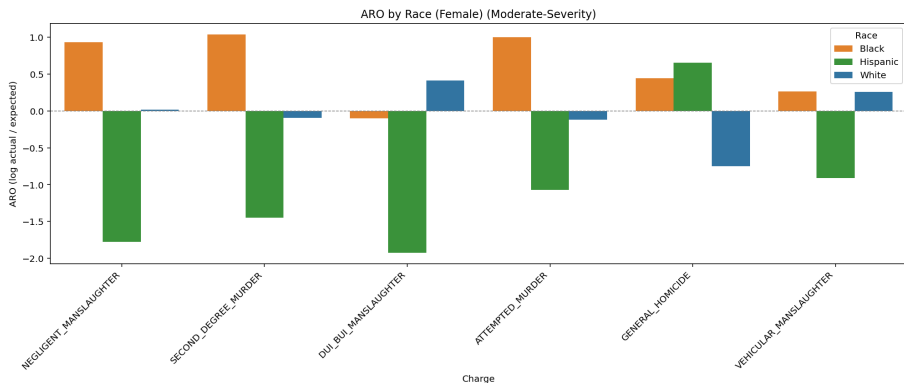
ARO Bar Plot — Female (All Charges)



ARO Bar Plot — Female (Severe Disparities)



ARO Bar Plot — Female (Moderate Disparities)



Simpson's Paradox and Sentencing Disparities

- **Goal:** Investigate whether observed disparities in sentencing severity (e.g., higher average sentence for one group) hold true across all subgroups — or if they reverse when broken down by charge type.
- **How ARO Helps:**
 - ARO identifies over/underrepresentation in each charge category.
 - It helps control for differences in the distribution of charges across groups.
- **How Average Sentence Helps:**
 - When combined with ARO, it shows if overrepresented groups also face longer sentences — or if severity claims are confounded by charge mix.
 - Allows for sentence comparison within the same offense category.
- **Simpson's Paradox Possibility:**
 - A group may have a higher overall average sentence...
 - ...but *lower* sentence within each individual charge category.
 - This reversal is a hallmark of Simpson's Paradox.

Using ARO and Sentence Length to Detect Simpson's Paradox

- **Step 1: Control for Charge Type**

- Use **ARO** to measure which groups are over- or underrepresented in each charge.
- ARO helps separate *who gets charged with what* from *how harshly they're sentenced*.

- **Step 2: Compare Sentence Lengths Within Charges**

- Analyze **average, median, and spread of sentence days** across demographic subgroups for each charge.
- This controls for the effect of charge severity.

- **Step 3: Identify Simpson's Paradox**

- Check if groups with **higher overall sentence averages** actually receive **lower or equal sentences** *within* specific charges.
- A paradox is present when the trend reverses once charges are accounted for.

- **Why This Matters:**

- Avoids misleading conclusions from aggregated data.

Thanks You

Thank You